

Figure 1 shows the relative error of the SAPT energy components versus the number of basis functions N . The main plot displays the error in kcal/mol for five different SAPT methods. The x-axis represents N (ranging from 2.0 to 4.8), and the y-axis represents the error in kcal/mol (ranging from -12 to 0). A vertical dashed line is drawn at $N=2.4$. The inset plot zooms in on the region where N is between 2.2 and 2.4 and the error is between -10 and -6 kcal/mol.

Legend:

- SAPT(PBE0)-D4 (S, ND) (Purple triangles)
- SAPT(PBE0)-D4 (I, ND) (Orange inverted triangles)
- SAPT(PBE0)-D4 (S) (Red squares)
- SAPT(PBE0)-D4 (I) (Green diamonds)
- SAPT2+3(CCD)/aTZ (Black circles)

Figure 1 shows the potential energy curve (PEC) for the reaction of a hydrogen atom with a water molecule. The x-axis represents the distance in Ångströms (Å), ranging from 2.0 to 5.0. The y-axis represents the energy in electronvolts (eV), ranging from 0.0 to 1.0. A vertical dashed line is drawn at 2.3 Å, indicating the transition state. The plot compares six different computational methods: SAPT(PBE0)-D4 (S) (red squares), SAPT(PBE0)-D4 (I) (green diamonds), HF-D4 (blue circles), SAPT(PBE0)/aDZ (brown crosses), SAPT(PBE0)/aTZ (grey plus signs), and SAPT2+3(CCD)/aTZ (black circles). The curves show a minimum around 2.3 Å, corresponding to the transition state, and converge at larger distances. The SAPT2+3(CCD)/aTZ method generally provides the highest energy values, while the SAPT(PBE0)-D4 (S) method provides the lowest values.